$y = e^{f(x)}$

$y' = e^{f(x)} \cdot f'(x)$

1] $y = e^{x}$

$y' = e^{x}$

2] $y = e^{3x}$

$y' = e^{3x} \cdot 3 = 3e^{3x}$

3] $y = e^{\sin x}$

$y' = e^{\sin x} \cdot \cos x \cdot 1$

4] $y = e^{\tan 2x}$

$y' = e^{\tan 2x} \cdot \sec^2 2x \cdot 2$

5] $y = e^{x \cos x}$

$y' = e^{x \cos x} \cdot \left[x \cdot (-\sin x) + \cos x \cdot 1 \right]$

6] $y = e^{\frac{x}{x+1}}$

$y' = e^{\frac{x}{x+1}} \left[\frac{(x+1) \cdot 1 - x \cdot 1}{(x+1)^2} \right]$

7] $y = e^{(5-\sin x)^2}$

$y' = e^{[5-\sin x]^2} \cdot 2 [5 - \sin x]' \cdot (-\cos x)$

8] $y = e^{\cot x} + \sin x - 4x^2 - 5x + 1$

$y' = e^{\cot x} \cdot (-\csc^2 x + 1) + \cos x \cdot 1 - 8x - 5$

9] $y = e^{x^2} \cdot \cos 2x$

$y' = e^{x^2} \cdot (-\sin 2x \cdot 2) +$

$\cos 2x \cdot (e^{x^2} \cdot 2x)$

10] $y = \dfrac{e^{3x} + 5x}{3x^2 + 1}$

$y' = \dfrac{(3x^2+1)\cdot[e^{3x} \cdot 3 + 5] - [e^{3x} + 5x][6x]}{(3x^2+1)^2}$

11] $y = (e^{\sec x} + 4x)^5$

$y' = 5(e^{\sec x} + 4x)^4 \cdot$

$(e^{\sec x} \cdot \sec x \tan x + 4)$

12] $y = e^{4x^2 - 5x}$

$y' = e^{4x^2-5x} \cdot (8x - 5)$

13] $y = e^{\sin 4x^3}$

$y' = e^{\sin 4x^3} \cdot \cos 4x^3 \cdot 12x^2$

14] $y = x^2 e^{(4x^4 - 5x)}$

$y' = x^2 \cdot e^{(4x^4-5x)} \cdot (16x^3 - 5) +$

$e^{4x^4-5x} \cdot 2x$

15) $x+y=x^2+y^2$

$1+y'=2x+2yy'$

$y'-2yy'=2x-1$

$y'(1-2y)=2x-1$

$$y'=\frac{2x-1}{1-2y}$$

16) $y+\sin x-\cos y+4x=0$

$y'+\cos x+\sin y\cdot y'+4=0$

$y'+\sin y\cdot y'=-\cos x-4$

$y'(1+\sin y)=-\cos x-4$

$$y'=\frac{-\cos x-4}{1+\sin y}$$

17) $e^y+\sin y-e^{2x}=y^2+3y^3-x^2$

$e^y\cdot y'+\cos y\cdot y'-2e^{2x}=2yy'+9y^2y'-2x$

$e^y\cdot y'+\cos y\cdot y'-2yy'-9y^2y'=-2x+2e^{2x}$

$y'[e^y+\cos y-2y-9y^2]=-2x+2e^{2x}$

$$y'=\frac{-2x+2e^{2x}}{e^y+\cos y-2y-9y^2}$$

18) $xy+y^2-5x=0$

$xy'+y+2yy'-5=0$

$xy'+2yy'=-y+5$

$y'(x+2y)=-y+5$

$$y'=\frac{-y+5}{x+2y}$$

19) $e^{xy}+\sin y+\cos x=1$

$e^{xy}(xy'+y)+\cos y\cdot y'-\sin x=0$

$e^{xy}\cdot xy'+e^{xy}\cdot y+\cos y\,y'-\sin x=0$

$e^{xy}\cdot xy'+\cos y\cdot y'=-e^{xy}\cdot y+\sin x$

$y'(e^{xy}\cdot x+\cos y)=-e^{xy}\cdot y+\sin x$

$$y'=\frac{-e^{xy}\cdot y+\sin x}{e^{xy}\cdot x+\cos y}$$

$y=\sqrt{f(x)}$

$$y'=\frac{f'(x)}{2\sqrt{f(x)}}$$

20) $\sqrt{x+y}=\sin x+\tan y$

$$\frac{1+y'}{2\sqrt{x+y}}=\cos x+\sec^2 y\cdot y'$$

$$\frac{1}{2\sqrt{x+y}}+\frac{y'}{2\sqrt{x+y}}=\cos x+\sec^2 y\cdot y'$$

$$\frac{y'\cdot 1}{2\sqrt{x+y}}-\sec^2 y\cdot y'=\cos x-\frac{1}{2\sqrt{x+y}}$$

$$y'\left[\frac{1}{2\sqrt{x+y}}-\sec^2 y\right]=\cos x-\frac{1}{2\sqrt{x+y}}$$

$$y'=\frac{\cos x-\frac{1}{2\sqrt{x+y}}}{\frac{1}{2\sqrt{x+y}}-\sec^2 y}$$

21 $5y - x^3 + 4y^2 = \sin x$

$5y' - 3x^2 + 8yy' = \cos x$

$5y' + 8yy' = \cos x + 3x^2$

$y'(5+8y) = \cos x + 3x^2$

$$y' = \frac{\cos x + 3x^2}{5+8y}$$

22 $y \sin x + \tan x = y^2$

$y\cos x + \sin x \cdot y' + \sec^2 x = 2yy'$

$\sin x\, y' - 2yy' = -y\cos x - \sec^2 x$

$y'(\sin x - 2y) = -y\cos x - \sec^2 x$

$$y' = \frac{-y\cos x - \sec^2 x}{\sin x - 2y}$$